GreenDA User Guide

GreenDA: GreenFeed® data analyzer

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Maintainer

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Brief description: This application is designed using R shiny to facilitate analysis of GreenFeed® (C-Lock Inc., Rapid City, SD) data. This application includes outlier elimination, diurnal variation adjustment, repeatability calculation, daily kinetic (average methane production and sum of number of visits), and descriptive statistical analysis of GreenFeed® data.

Acknowledgement: This work was supported by the Methane 2030 project (France).
All scripts may be freely modified and used. If you publish data analyzed using this program in any format (newsletter, academic article, etc.), you must cite the technical note (Jeon et al., 2026) for this program as a reference.

01 Data files preparation

1. GreenFeed data file

- There is no need to change the format, but each column name only allows English alphabets and simple special characters.
- The date format is encoded based on UTC (i.e. origin = "1970-01-01", tz = "UTC").
- The decimal point of the number must be separated by a period. Commas are not allowed.

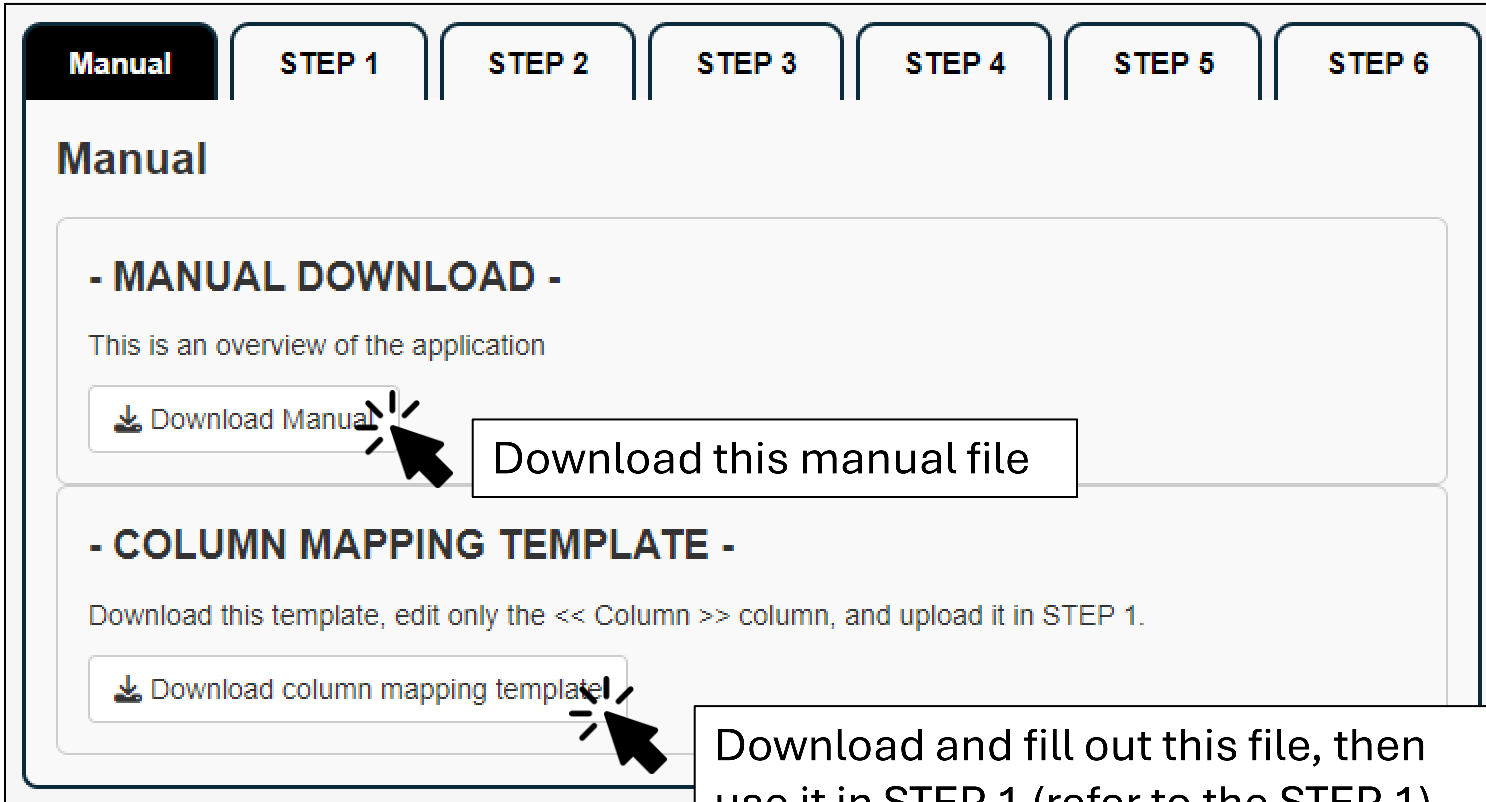
2. Factor file

- The Factor file must use the common ID as the Animal ID in the GreenFeed data file.
- There must be only one line of data per animal.

3. General

- All steps must be performed in order (Step 1 → Step 6).
- If you wish to skip, you should click the « Skip this step » button for each step.

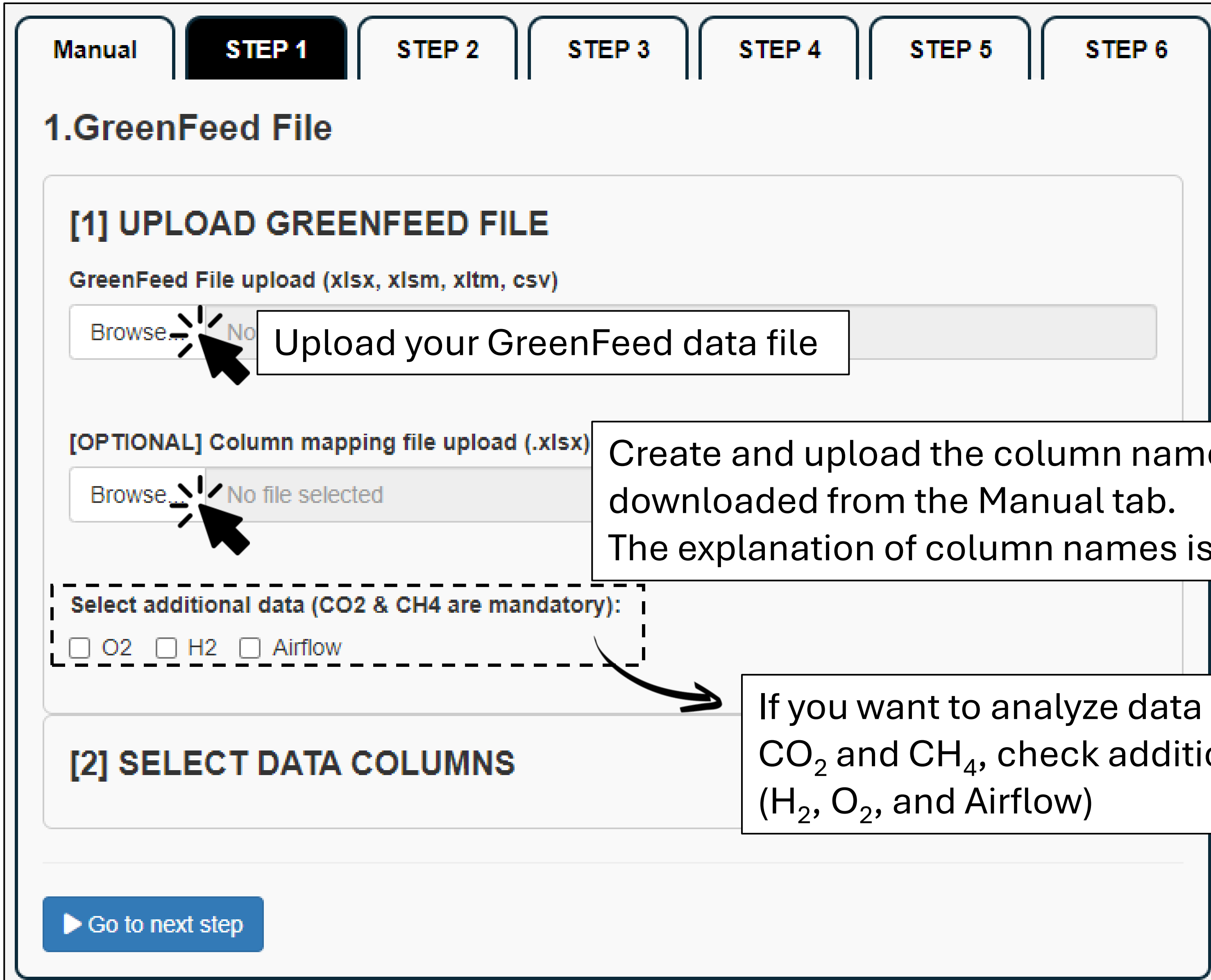
02 [Manual] Manual & Column name matching file



Download this manual file

Download and fill out this file, then use it in STEP 1 (refer to the STEP 1).

02 [STEP1] GreenFeed data file upload



Upload your GreenFeed data file

Create and upload the column name mapping file downloaded from the Manual tab.
The explanation of column names is in the next page.

If you want to analyze data other than CO_2 and CH_4, check additional data (H_2, O_2, and Airflow)

02 [STEP1] GreenFeed data file upload

*How to prepare a « **Column Mapping Template** » ?*

Field	Column
animal	Farm Name
machine	FID
startdt	Start Time
enddt	End Time
duree_day	Good Data Duration
hour_day	Hour Of Day
CO2	CO2 Massflow (g/d)
CH4	CH4 Massflow (g/d)
O2	O2 Massflow (g/d)
H2	H2 Massflow (g/d)
airflow_col	Average Airflow (L/s)

/!\ *Gray cells cannot be changed*

Animal ID column name

Machine ID column name

Start time column name

End time column name

Stay duration column name

Decimal hours (e.g. 16h30 → 16.5) column name

CO_2 (g/d) column name

CH_4 (g/d) column name

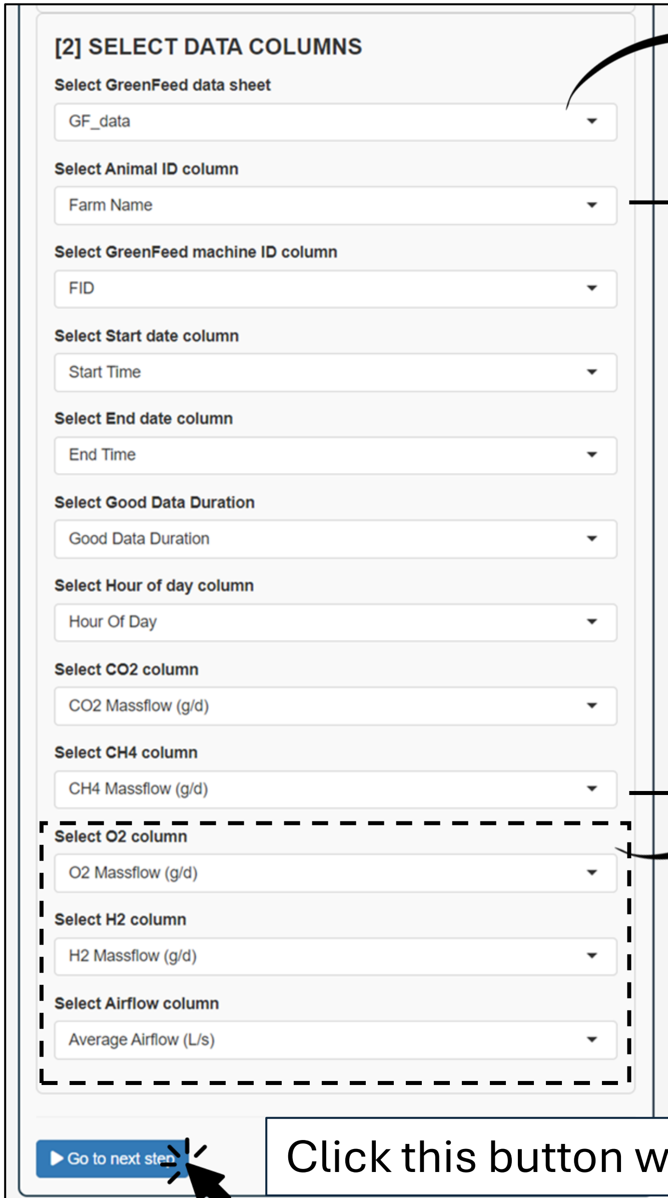
O_2 (g/d) column name

H_2 (g/d) column name

Airflow (L/s) column name

If these data are not available, leave them blank

02 [STEP1] GreenFeed data file upload



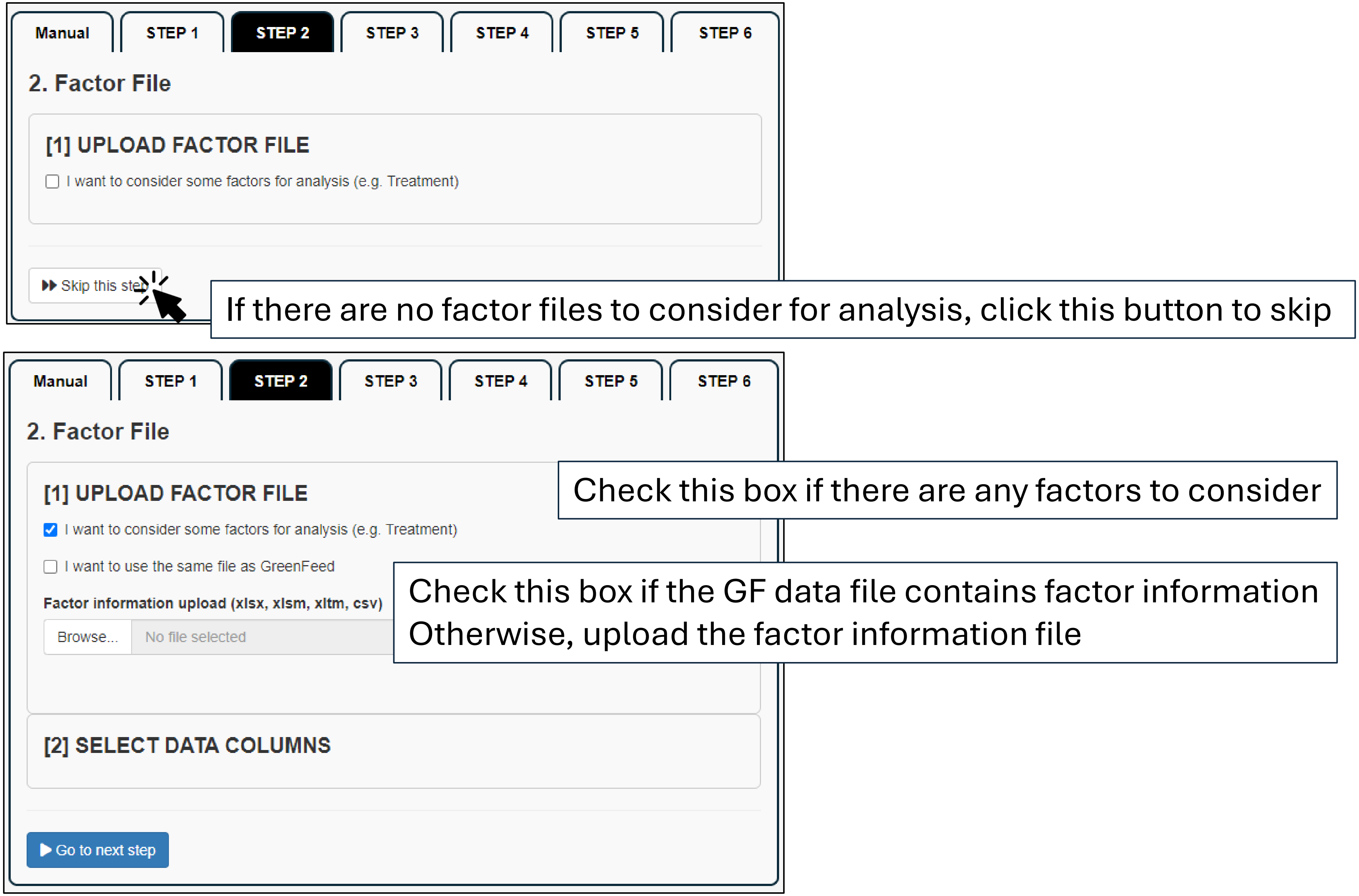
The name of the data sheet

Select each column name of your data file

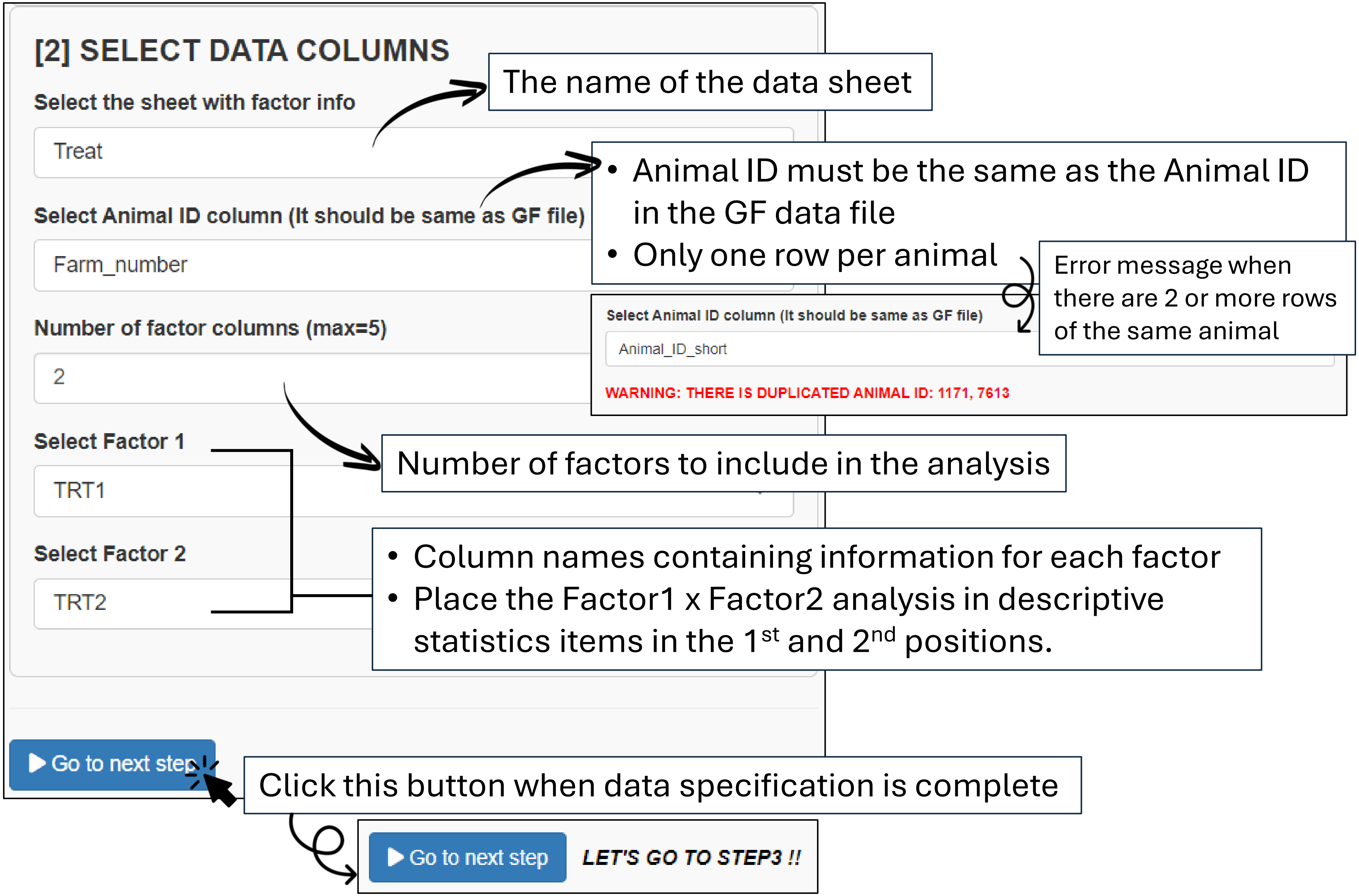
Activate only
when O_2, H_2, or Airflow data is available

Click this button when data specification is complete

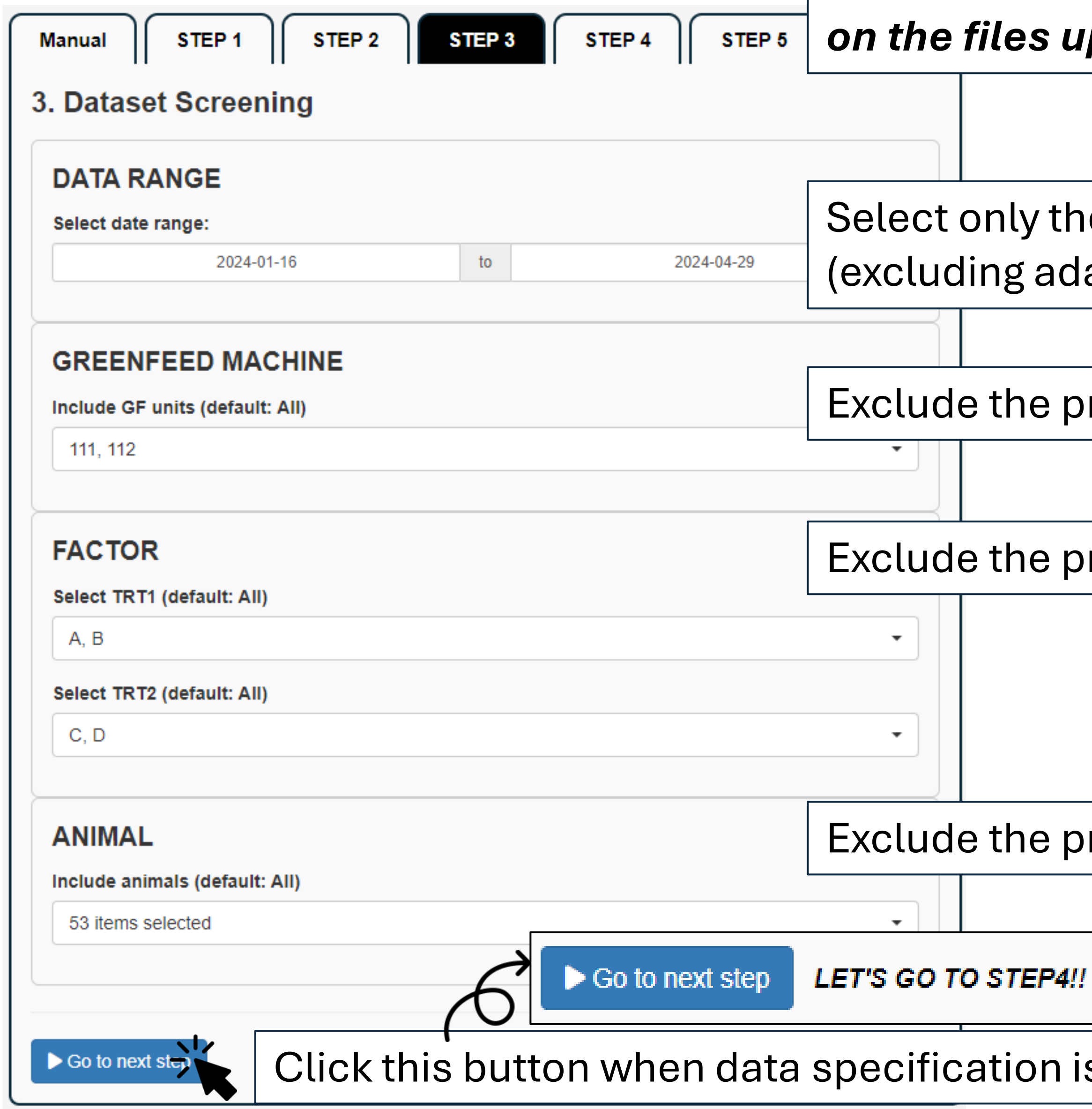
02 [STEP2] Factor file upload



02 [STEP2] Factor file upload



02 [STEP3] Dataset Screening



This tab automatically generated based on the files uploaded by the user

Select only the actual data collection period (excluding adaptation and transition period)

Exclude the problematic machine

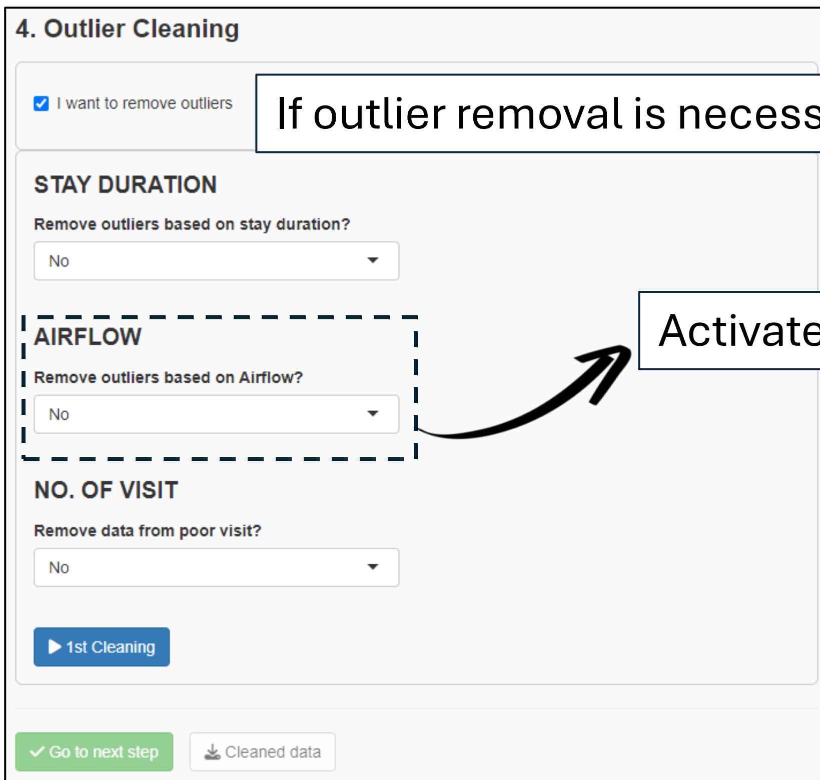
Exclude the problematic factor

Exclude the problematic animal

Click this button when data specification is complete

02 [STEP4] Outlier cleaning

If you have already removed outliers, click the button to skip this step

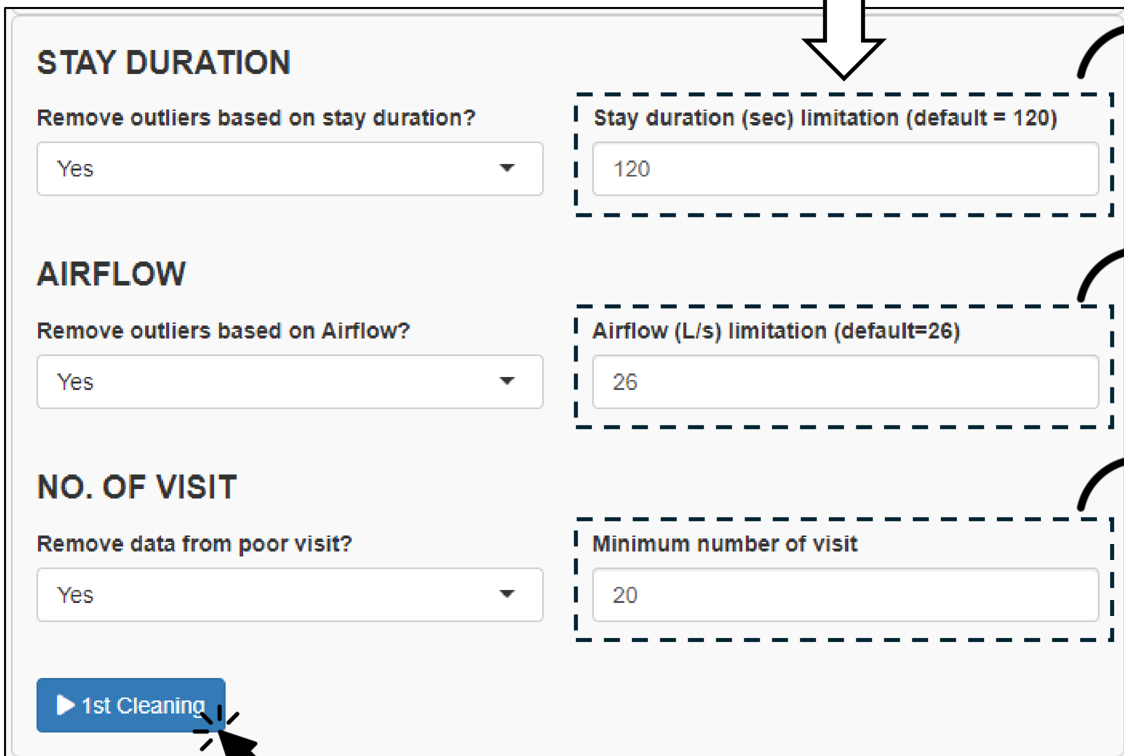


If outlier removal is necessary, select this checkbox to start outlier removal

Activate only when Airflow data is available

02 [STEP4] Outlier cleaning

Active only if the left item is « Yes »



Delete data line if stay duration is less than this value

Delete data line if airflow is less than this value

Delete data lines for animals with fewer than this number of visits throughout the entire period

Click this button when data specification is complete

02 [STEP4] Outlier cleaning

Data lines with problematic CO_2 or CH_4 values are removed

[1] Skip CO_2 & CH_4 cleaning

STATISTICALLY - CO2 & CH4

Select a method to remove CO2, CH4 outlier

No

▶ 2nd Cleaning

If cleaning of CO_2 and CH_4 is unnecessary, select « No » for method and click the button

[2] Correlation based method

STATISTICALLY - CO2 & CH4

Select a method to remove CO2, CH4 outlier

Arbre_2016

R² calculation

Click the « R^2 calculation » button to check the R^2 and *P-value*

R² calculation

Items	Value
n	19716
R	0.69
R²	0.47
P-value	< 2.22e-16
Significant	Yes

▶ 2nd Cleaning

Click this button for cleaning

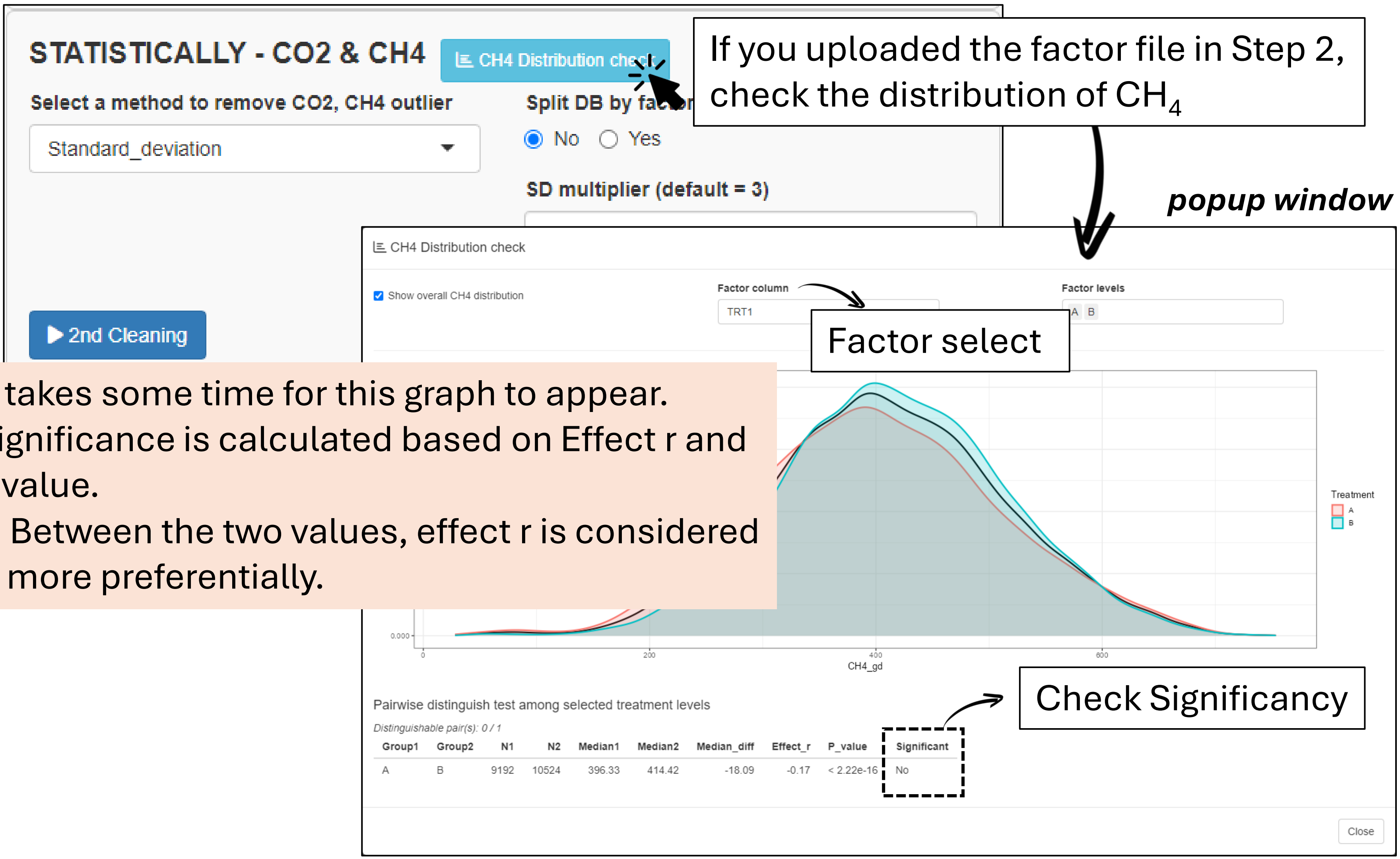
02 [STEP4] Outlier cleaning

Data lines with problematic CO_2 or CH_4 values are removed

[3] Standard deviation based method

If you uploaded the factor file in Step 2, check the distribution of CH_4

popup window



Group1	Group2	N1	N2	Median1	Median2	Median_diff	Effect_r	P_value	Significant
A	B	9192	10524	396.33	414.42	-18.09	-0.17	< 2.22e-16	No

Factor select

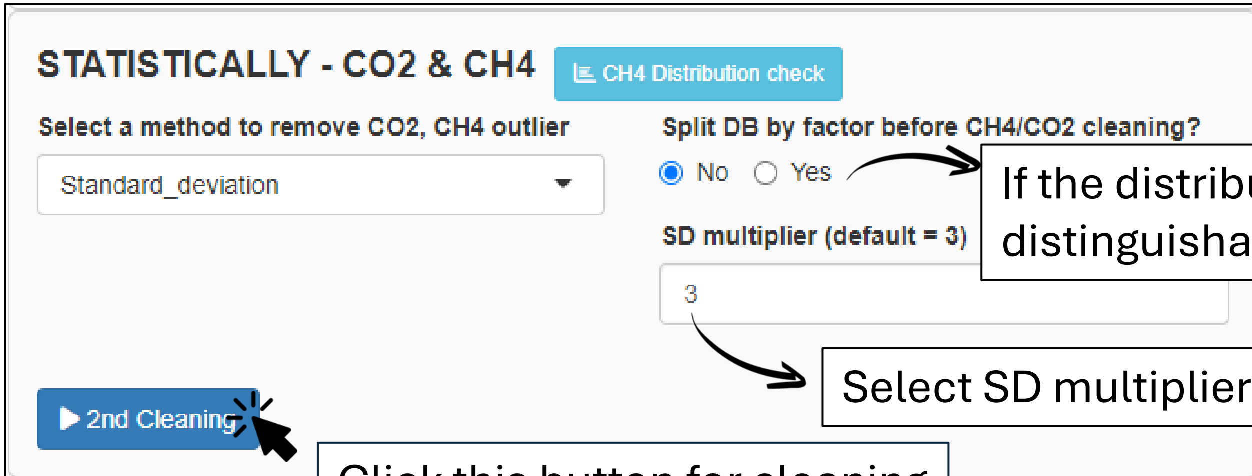
Check Significancy

* It takes some time for this graph to appear.
* Significance is calculated based on Effect r and P-value.
→ Between the two values, effect r is considered more preferentially.

02 [STEP4] Outlier cleaning

Data lines with problematic CO_2 or CH_4 values are removed

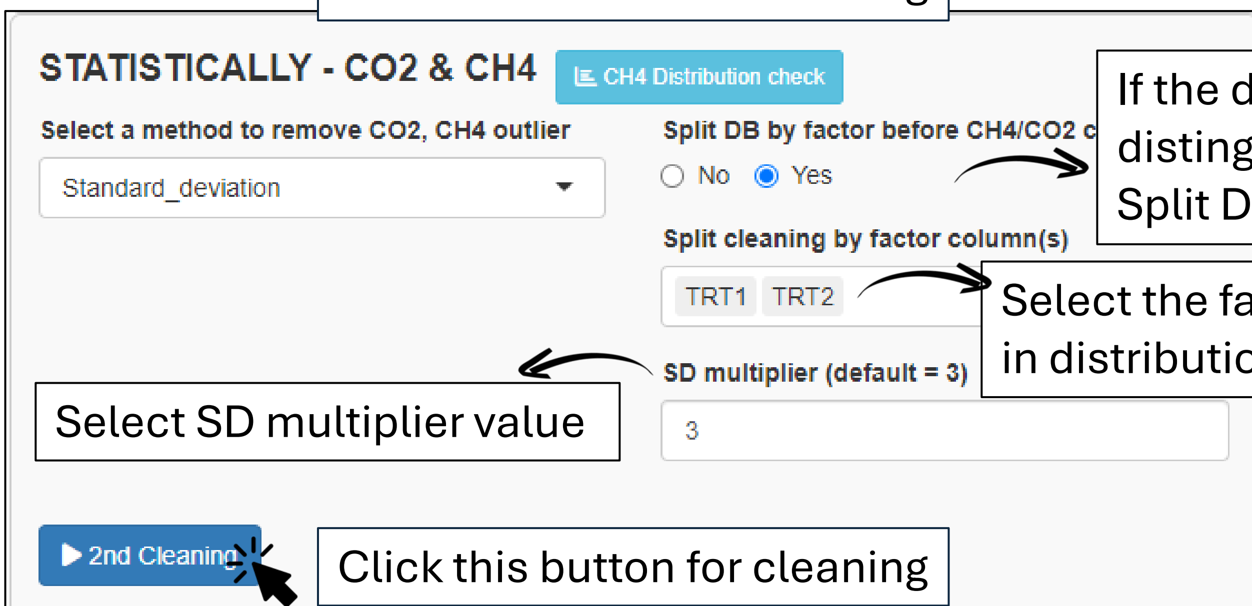
[3] Standard deviation based method



If the distribution is not significantly distinguishable, select No for Split DB

Select SD multiplier value

Click this button for cleaning



If the distribution is significantly distinguishable, select Yes for Split DB

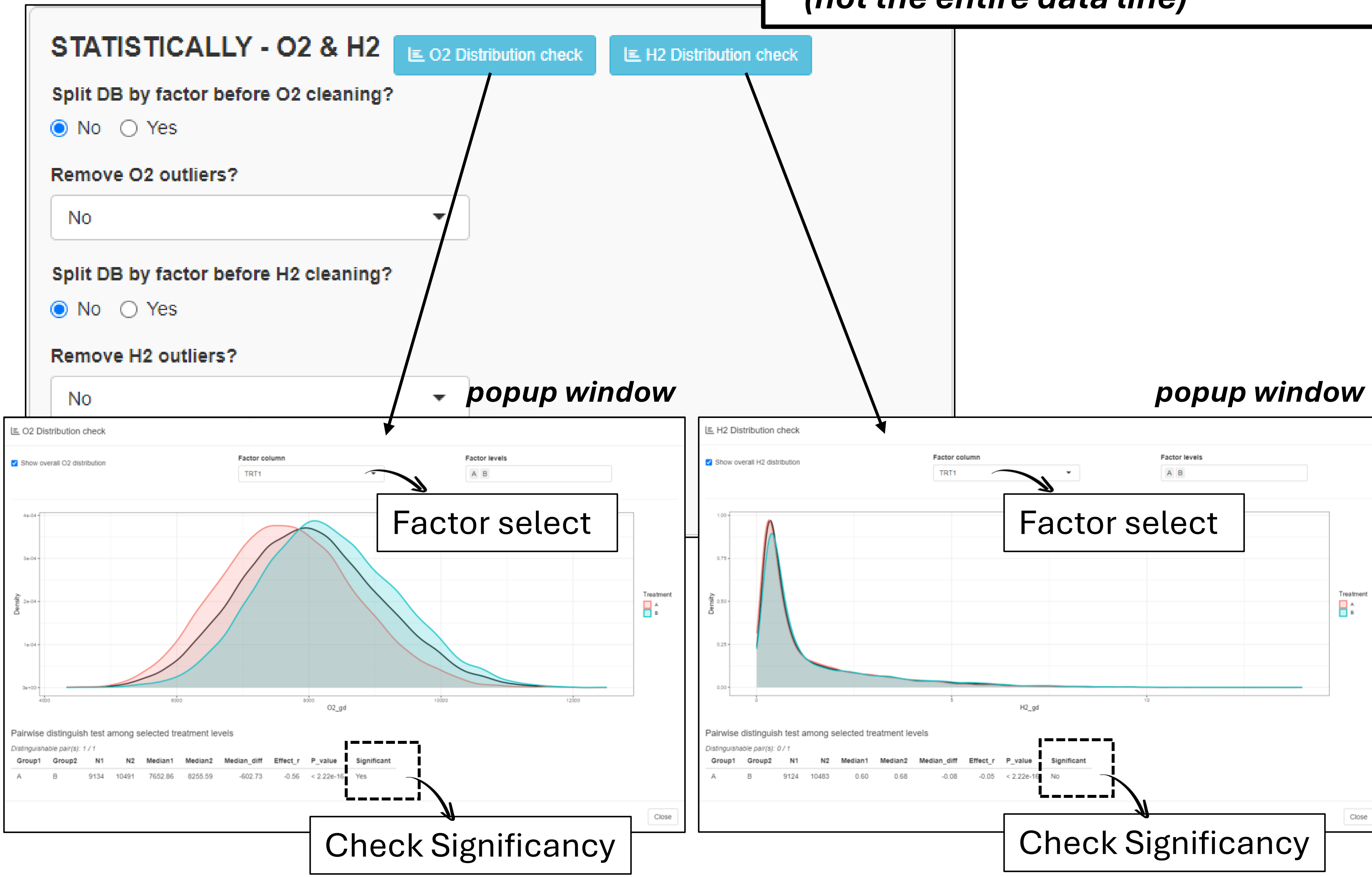
Select the factor where the distinction in distribution was significant

Select SD multiplier value

Click this button for cleaning

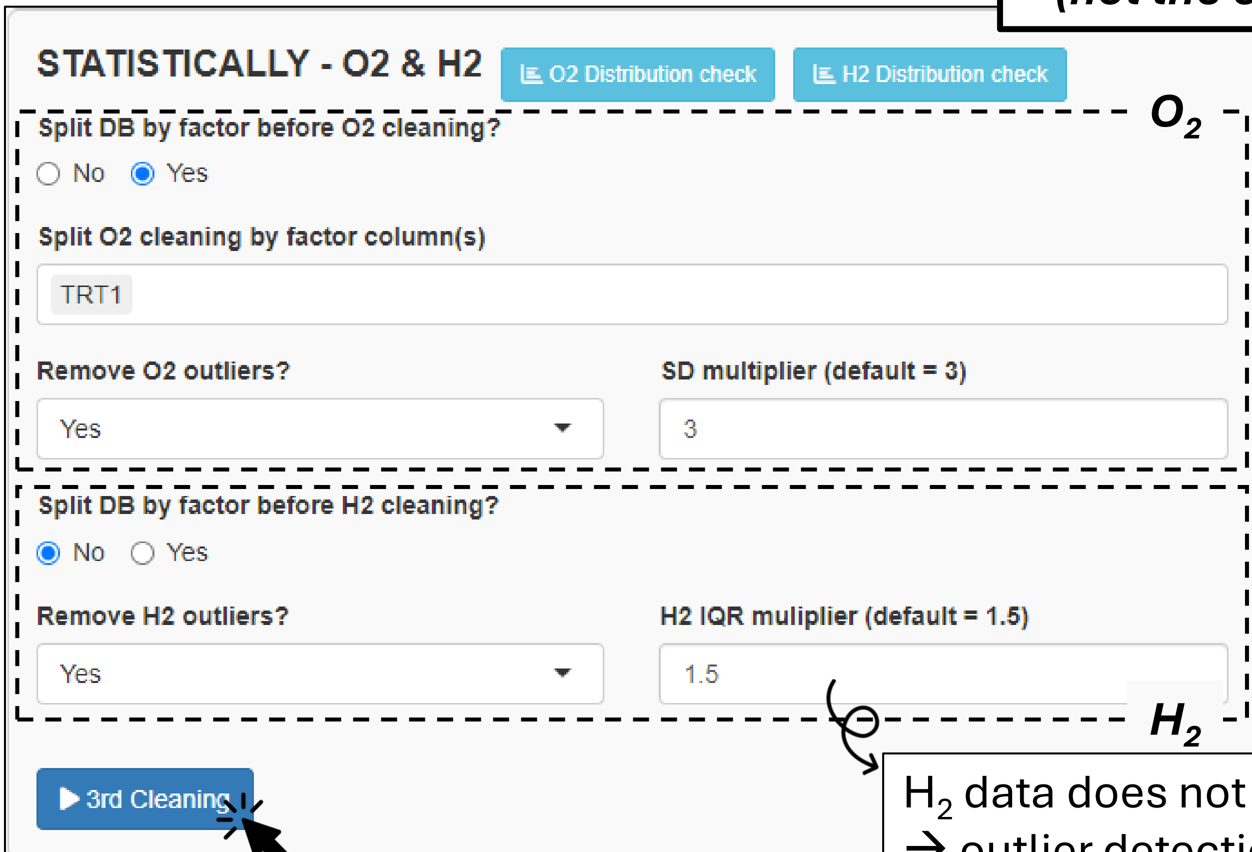
02 [STEP4] Outlier cleaning

- ***Activate only if O_2 or H_2 values exist***
- ***Delete only the problematic value (not the entire data line)***



02 [STEP4] Outlier cleaning

- ***Activate only if O_2 or H_2 values exist***
- ***Delete only the problematic value (not the entire data line)***



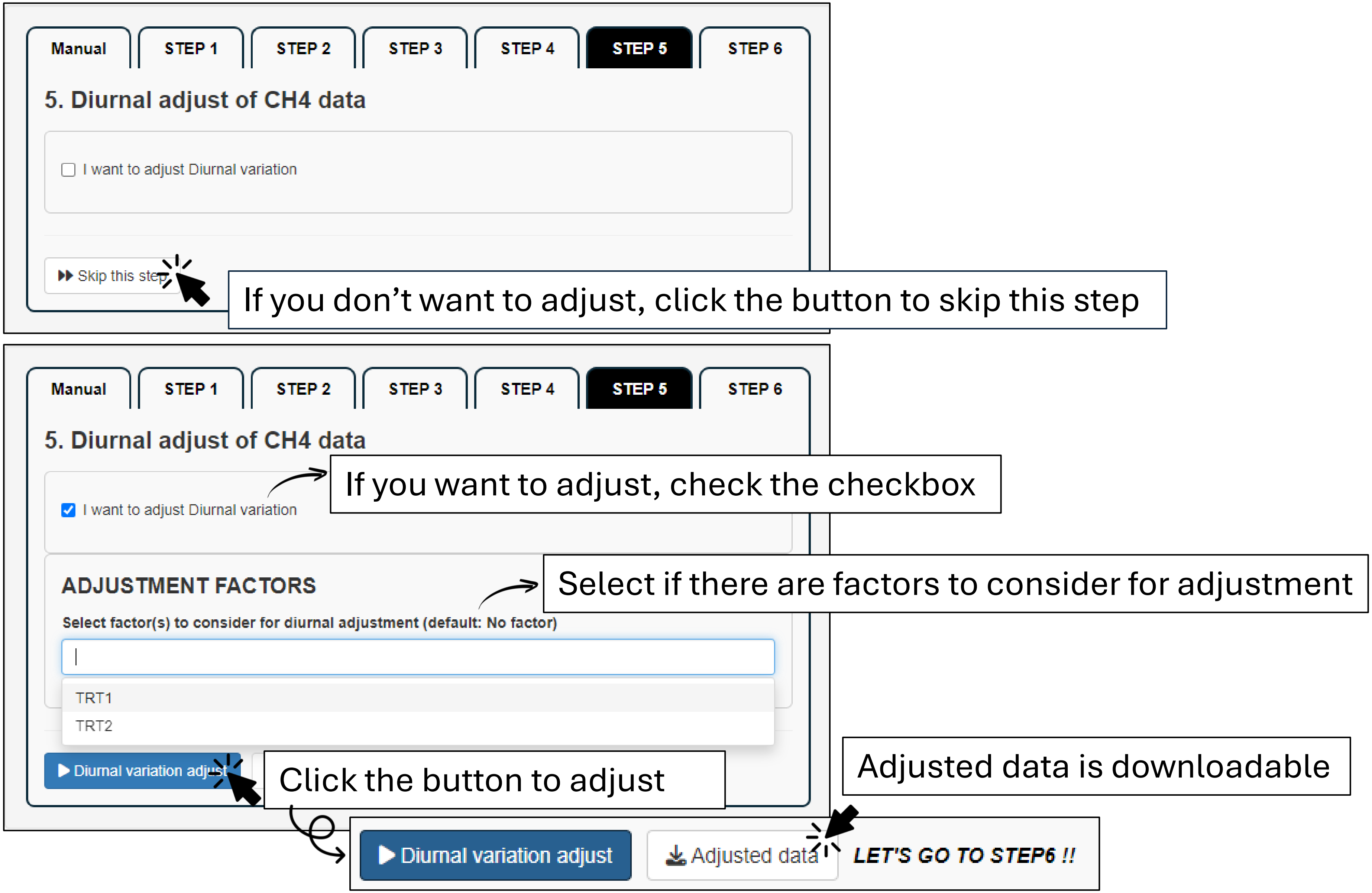
H_2 data does not follow a normal distribution → outlier detection based on IQR is used.

Click this button for cleaning

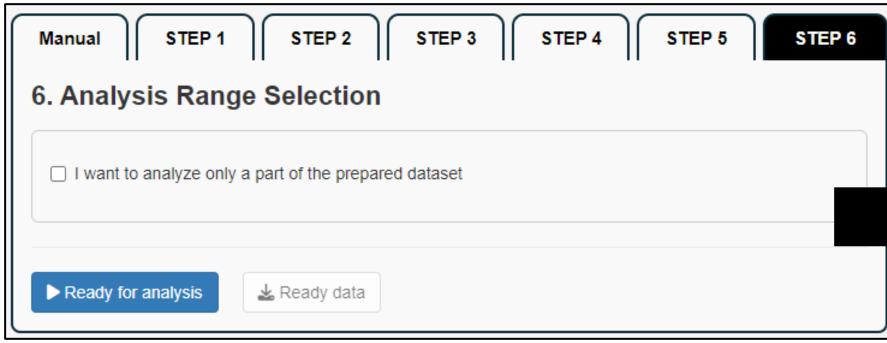
Click this button to check the number of outliers and proceed to the next step

Cleaned data is downloadable

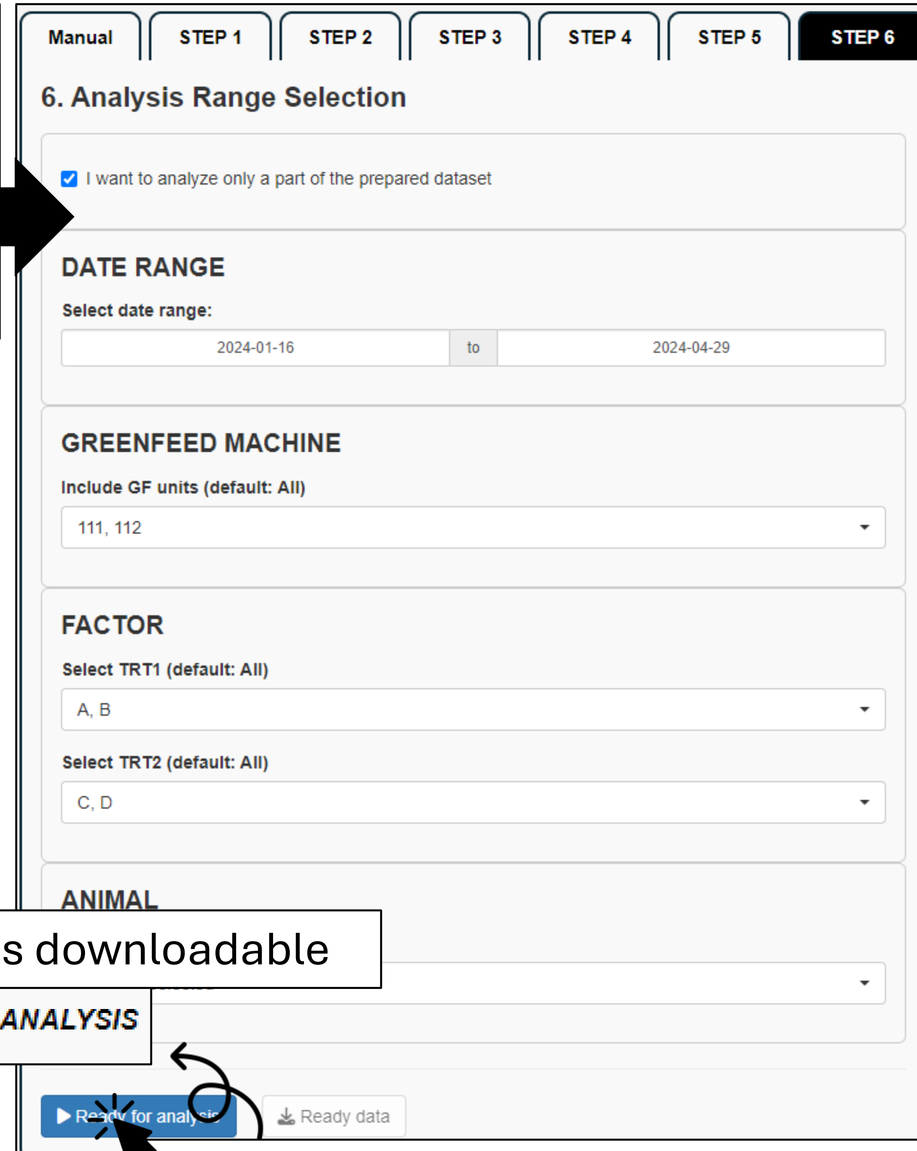
02 [STEP5] Diurnal adjust of CH_4 data

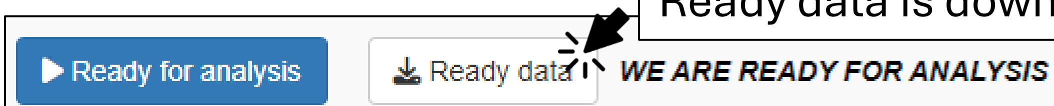


02 [STEP6] Analysis range selection



If you want to analyze only a part of the data, specify the date range, machine, factor, or animal you wish to analyze

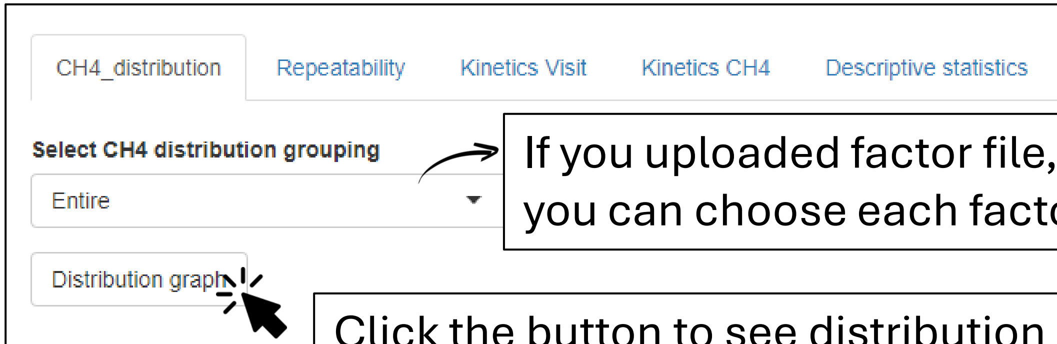




Ready data is downloadable

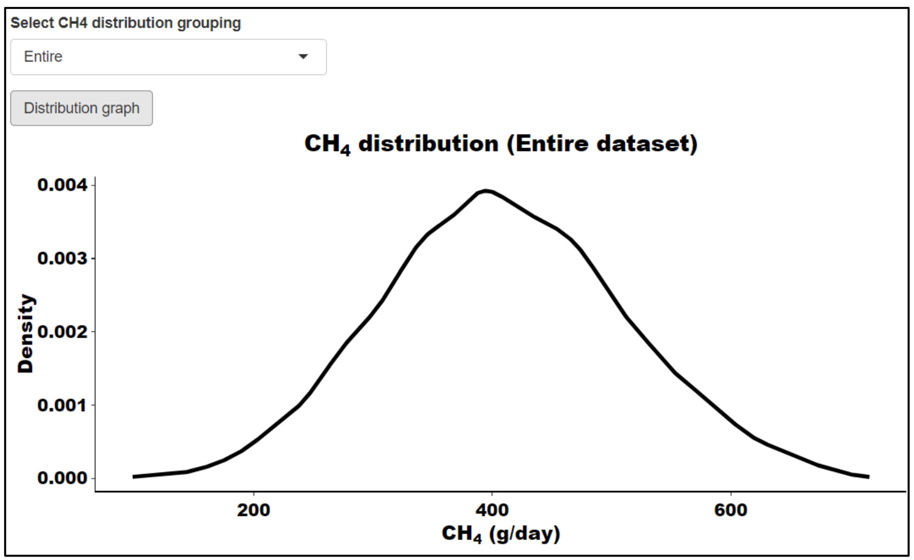
Click the button to ready for analysis

03 Methane distribution graph

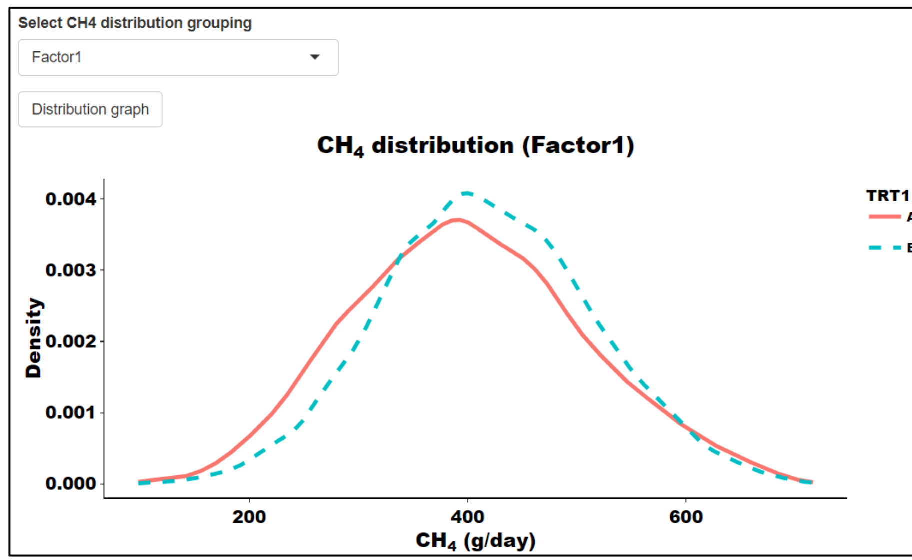


If you uploaded factor file,
you can choose each factor as well

Click the button to see distribution

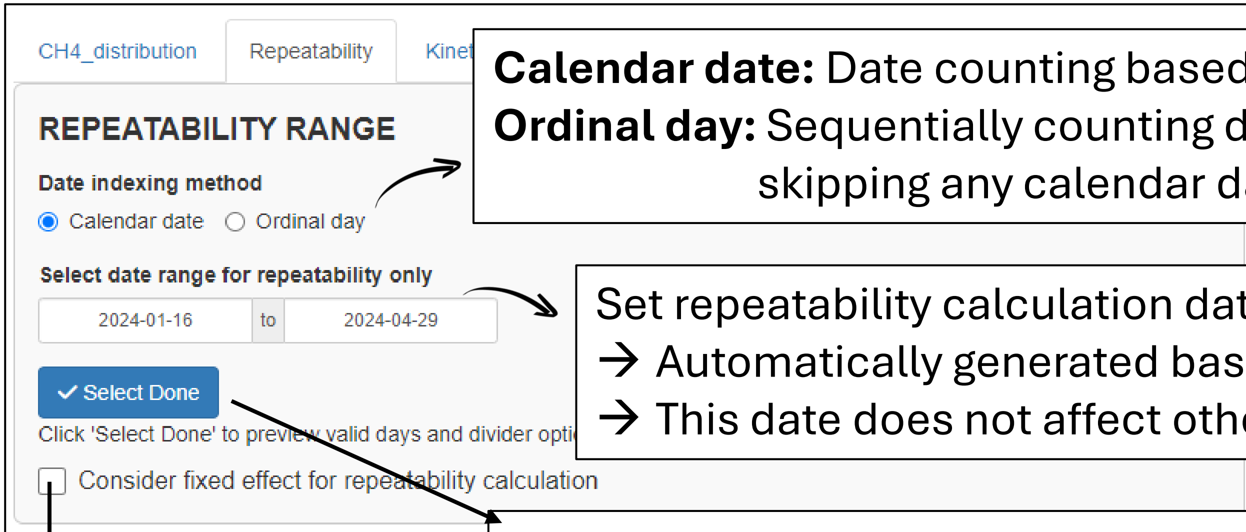


e.g.) Entire dataset distribution



e.g.) Factor1 distribution

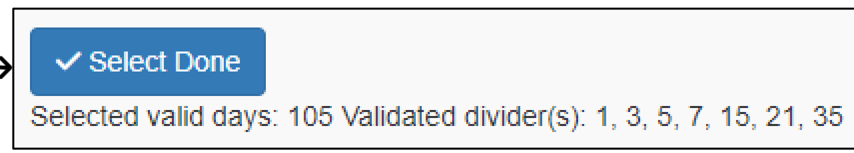
03 Repeatability calculation



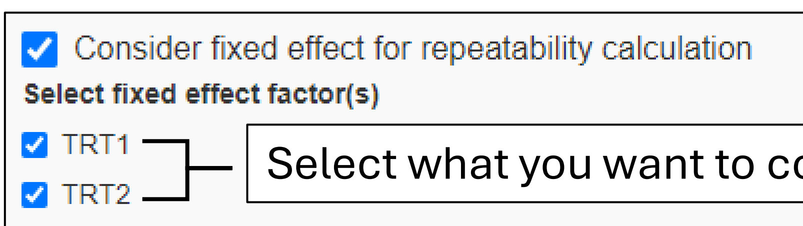
Calendar date: Date counting based on calendar date
Ordinal day: Sequentially counting days for each animal, skipping any calendar days with no recorded date

Set repeatability calculation date range
→ Automatically generated based on « Ready data »
→ This date does not affect other analyses

Click this button to check total number of days & validated dividers

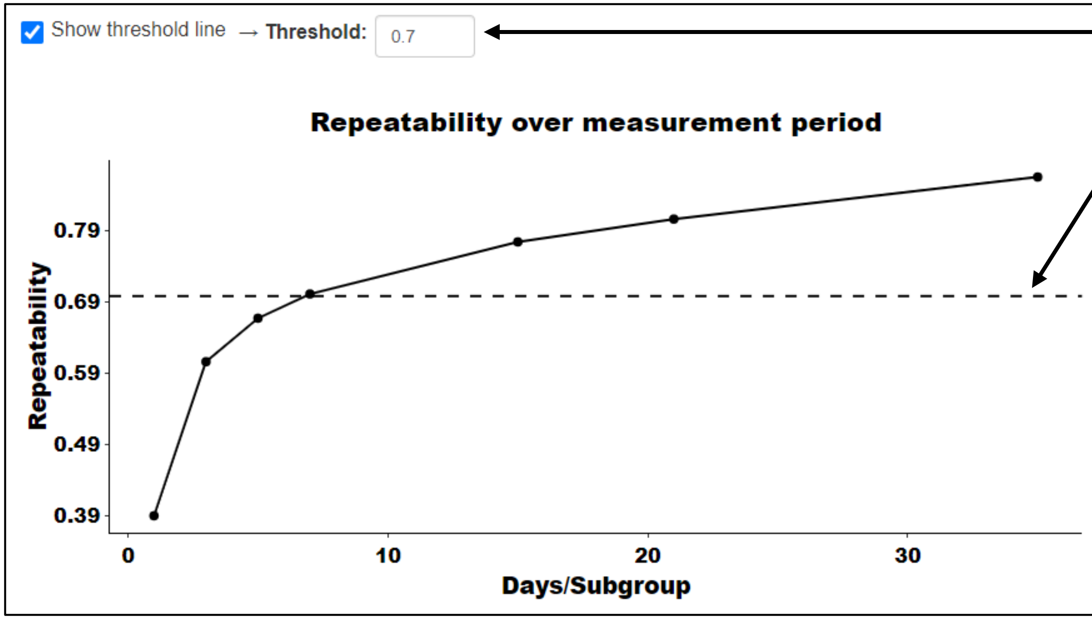


Check it if you want to consider factors for repeatability calculation

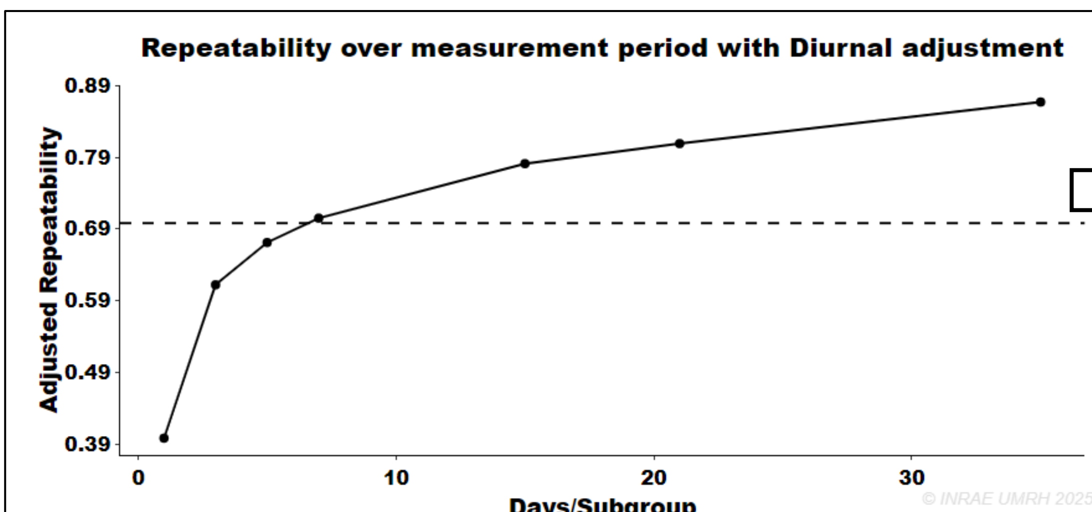


Select what you want to consider

03 Repeatability calculation

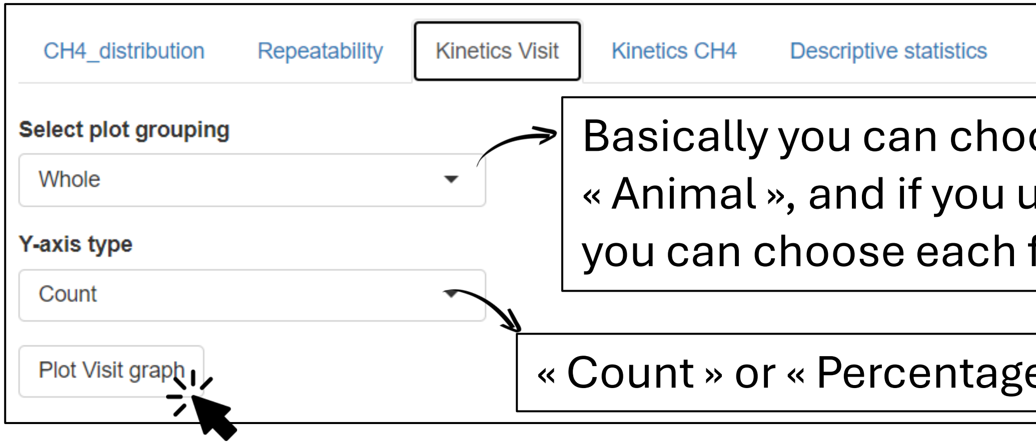


User can set their own threshold line



This graph is displayed only if the user adjusted the diurnal variation in Step 5

03 Daily kinetics – Number of visits

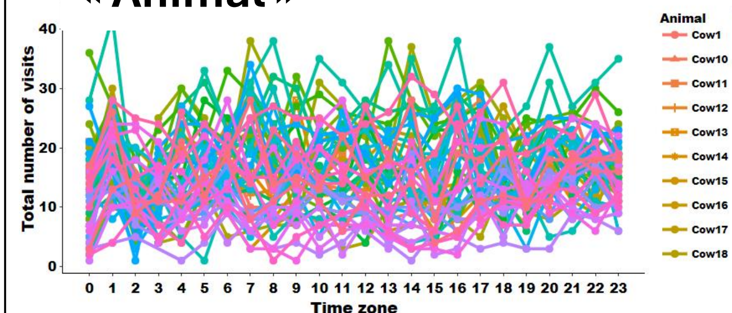


Basically you can choose « Whole » or « Animal », and if you uploaded factor file, you can choose each factor as well

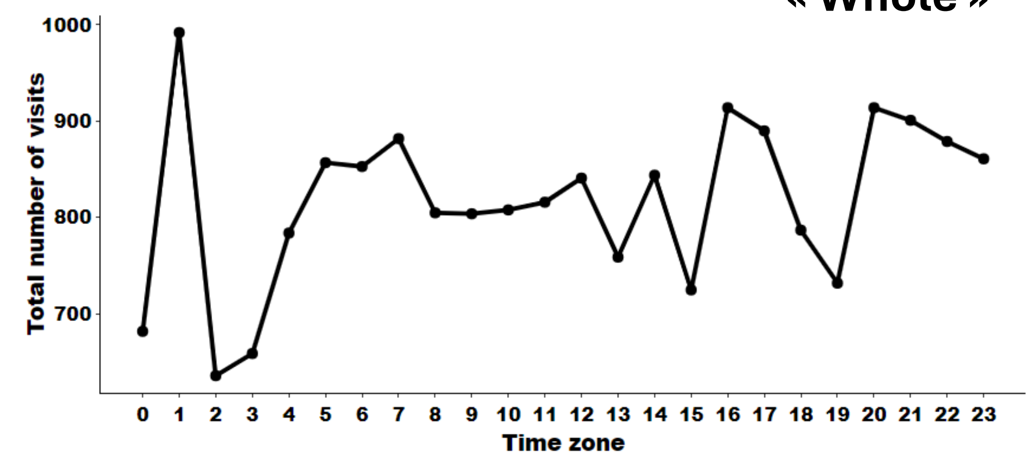
« Count » or « Percentage »

Click the button to see distribution

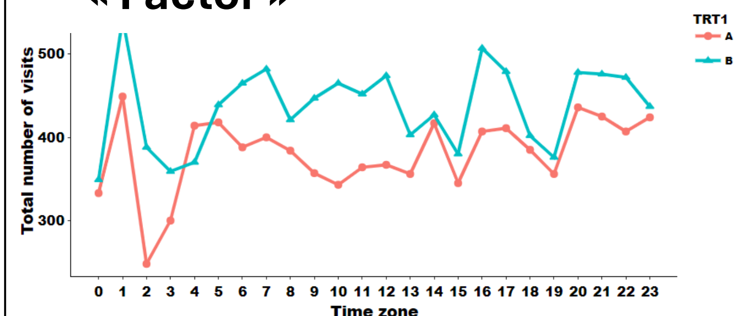
« Animal »



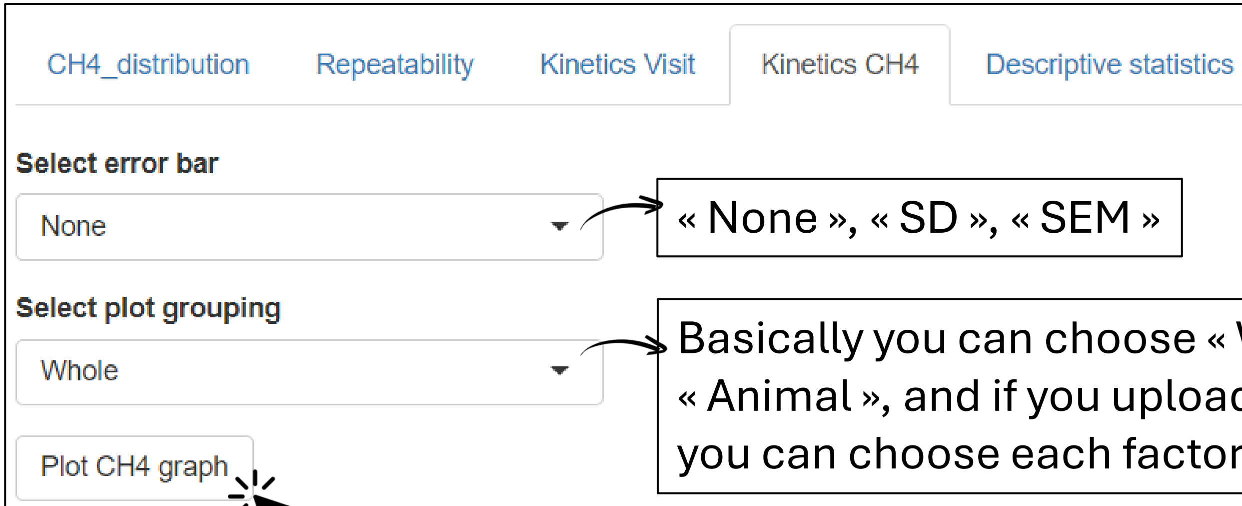
« Whole »



« Factor »



03 Daily kinetics – CH_4 production

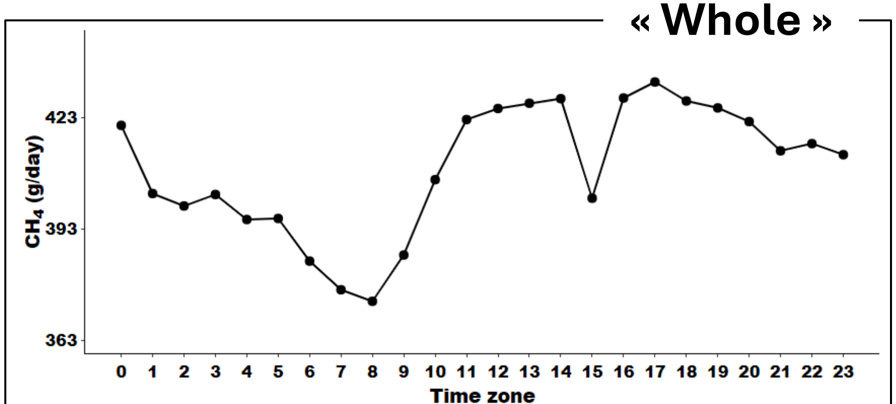


« None », « SD », « SEM »

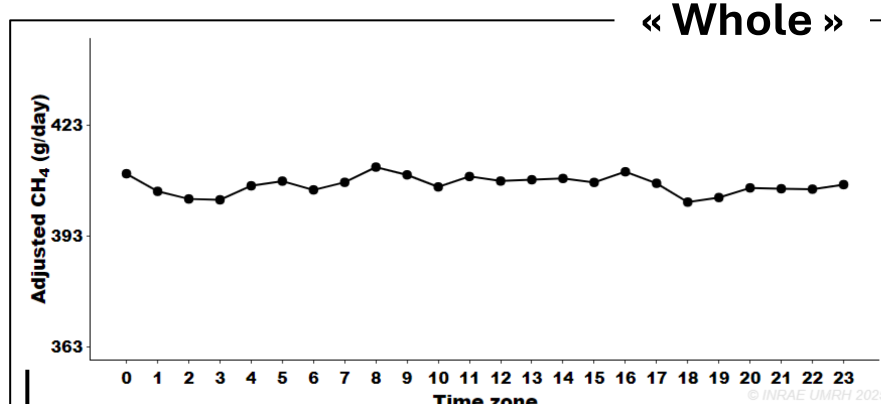
Basically you can choose « Whole » or « Animal », and if you uploaded factor file, you can choose each factor as well

Click the button to see distribution

« Whole »

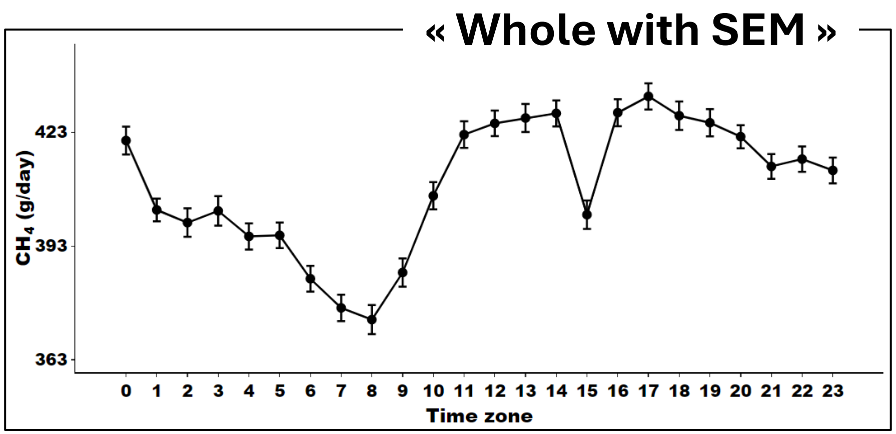


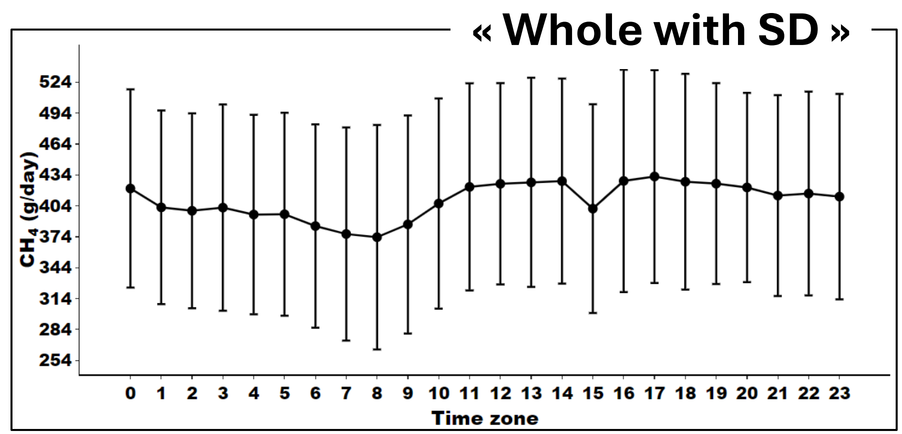
« Whole »

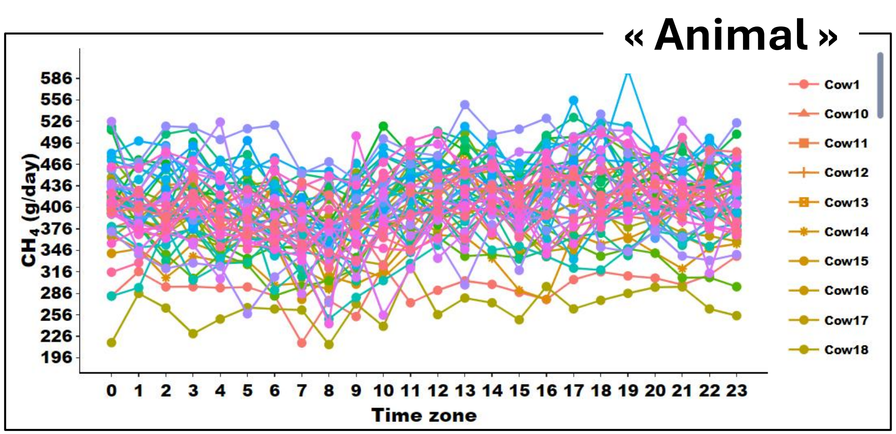


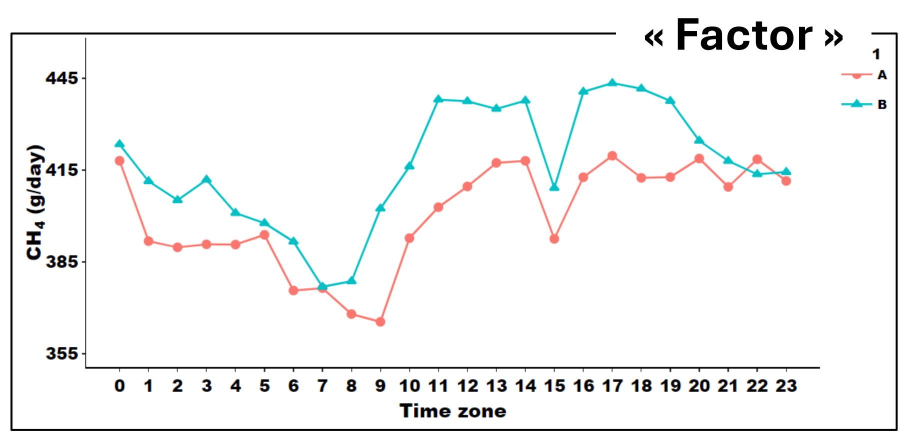
This graph is displayed only if the user adjusted the diurnal variation in Step 5

03 Daily kinetics – CH_4 production

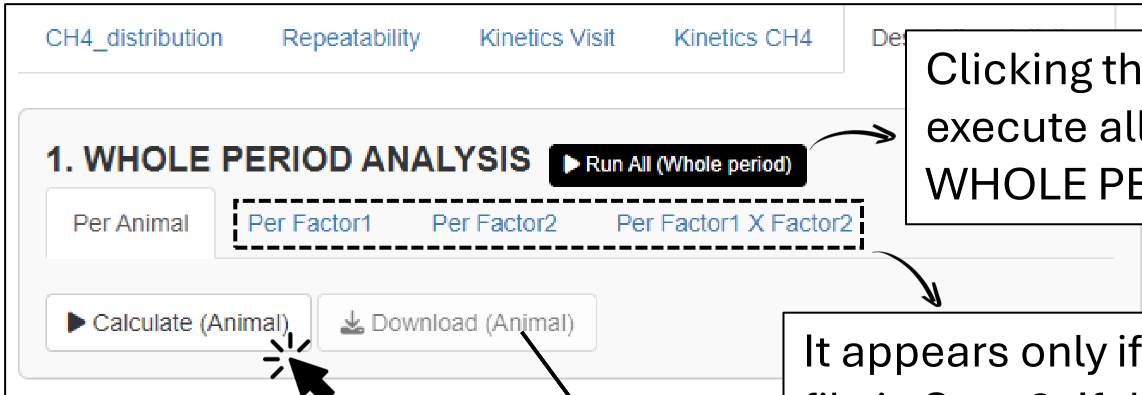








03 Descriptive statistics



Clicking this button will automatically execute all « Calculate » buttons in the WHOLE PERIOD ANALYSIS session

It appears only if you upload and specify a factor file in Step 2. If there are more than 2 factors, Factor1 x Factor2 is also enabled

Click the button to calculate

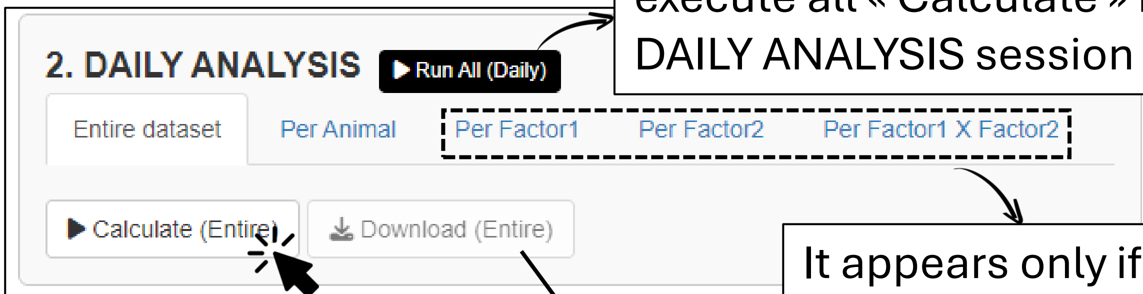
Download descriptive statistic results

This includes daily descriptive statistics:

- Total number of visits
- Total number of animals
- Total number of visits per animal
- Average of daily gas emission & standard deviation
- (If user requested diurnal adjustment) All the results of CH_4 after diurnal adjustment

03 Descriptive statistics

Clicking this button will automatically execute all « Calculate » buttons in the DAILY ANALYSIS session



It appears only if you upload and specify a factor file in Step 2. If there are more than 2 factors, Factor1 x Factor2 is also enabled

Click the button to calculate

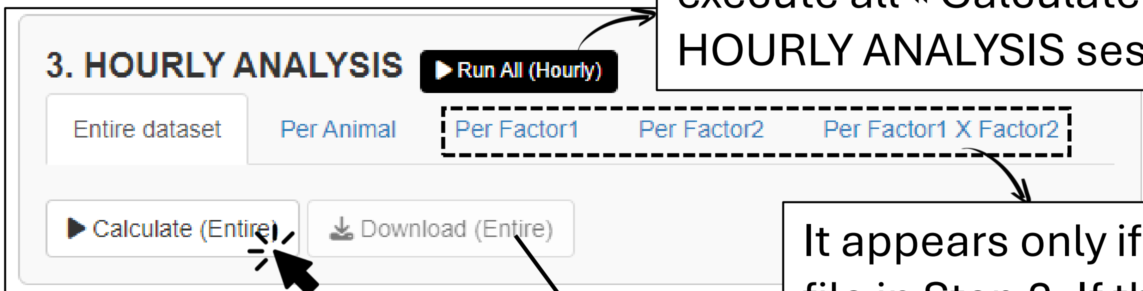
Download descriptive statistic results

This includes daily descriptive statistics:

- Daily number of visits
- Daily number of animals
- Daily number of visits per animal
- Average of daily gas emission & standard deviation
- (If user requested diurnal adjustment) All the results of CH_4 after diurnal adjustment

03 Descriptive statistics

Clicking this button will automatically execute all « Calculate » buttons in the HOURLY ANALYSIS session



It appears only if you upload and specify a factor file in Step 2. If there are more than 2 factors, Factor1 x Factor2 is also enabled

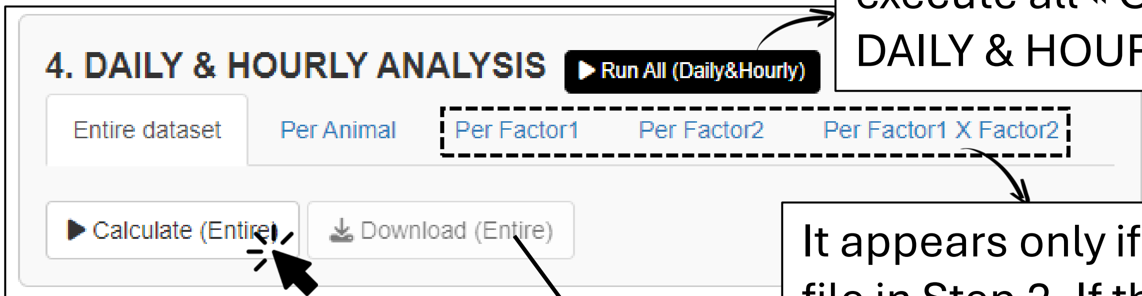
Click the button to calculate

Download descriptive statistic results

This includes daily descriptive statistics:

- Sum of stay duration per hour
- Sum of number of visits per hour
- Number of animals per hour
- Number of visits per animal per hour
- Average gas emission & standard deviation per hour
- (If user requested diurnal adjustment) All the results of CH_4 after diurnal adjustment

03 Descriptive statistics



Clicking this button will automatically execute all « Calculate » buttons in the DAILY & HOURLY ANALYSIS session

It appears only if you upload and specify a factor file in Step 2. If there are more than 2 factors, Factor1 x Factor2 is also enabled

Click the button to calculate

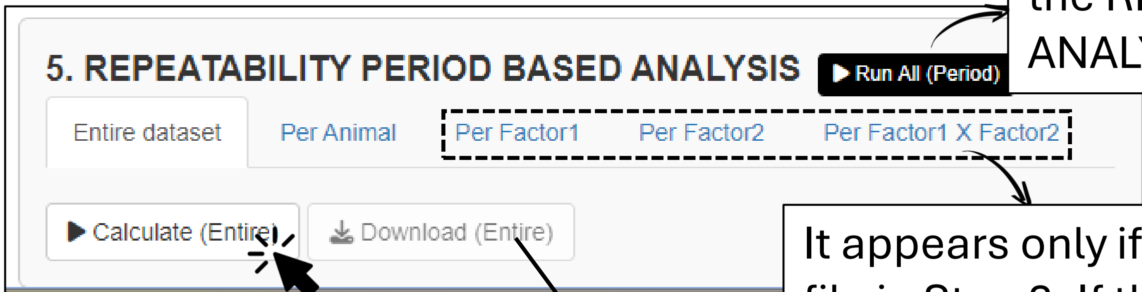
Download descriptive statistic results

This includes daily descriptive statistics:

- Sum of stay duration per hour within a day
- Sum of number of visits per hour within a day
- Number of animals per hour within a day
- Number of visits per animal per hour within a day
- Average gas emission & standard deviation per hour within a day
- (If user requested diurnal adjustment) All the results of CH_4 after diurnal adjustment

03 Descriptive statistics

Clicking this button will automatically execute all « Calculate » buttons in the REPEATABILITY PERIOD BASED ANALYSIS session



It appears only if you upload and specify a factor file in Step 2. If there are more than 2 factors, Factor1 x Factor2 is also enabled

Click the button to calculate

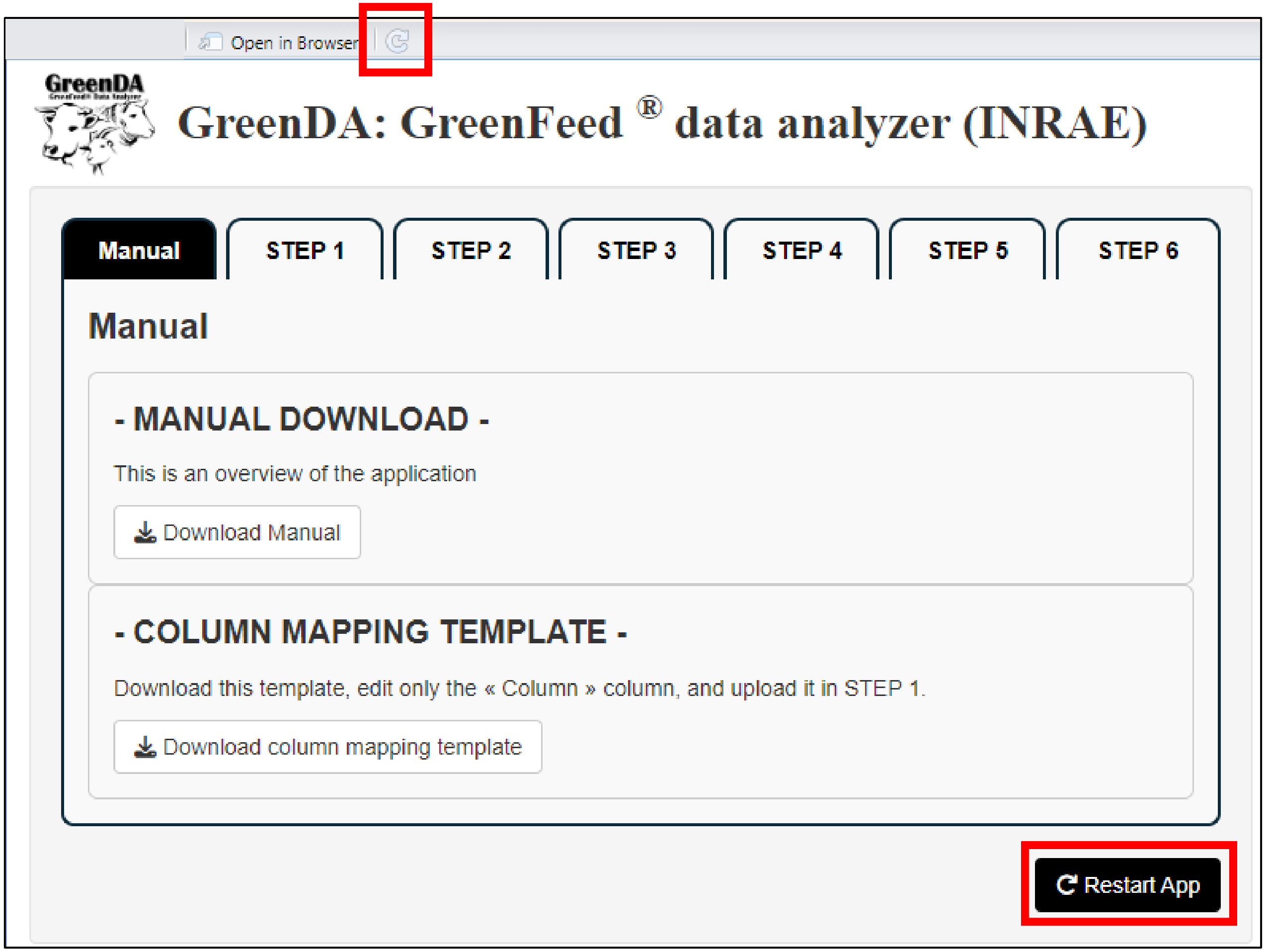
Download descriptive statistic results

This includes daily descriptive statistics:

This is possible only after repeatability calculation is performed first

- Repeatability results
- Sum of number of visits per subgroup period
- Number of animals per subgroup period
- Number of visits per animal per subgroup period
- Average gas emission & standard deviation per subgroup period
- (If user requested diurnal adjustment) All the results of CH_4 after diurnal adjustment

04 Application Restart



* Restarting the app is highly recommended when uploading a new dataset.
* In this case, please use the « Restart App » button at the bottom of each Step, rather than the ↻ button at the top of the window.